

Local Floer Chain Complex across Hamiltonian Bifurcation

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This talk is based on the ongoing work with Hong-kwon Jo (SNU).

- **Motivation and Preliminaries.**

Bifurcation, Conley–Zehnder index and Floer homology.

- **Classification of Hamiltonian bifurcations.**

Birth-death, period doubling, phantom kiss and emission.

- **Chain complex mutation.**

Construction of cylinders, Morse–Bott cascade counting.

Main result. Describing the explicit change of Conley–Zehnder indices and Floer chain complex for generic Hamiltonian bifurcation types.

Motivation

Motivation 1. Bifurcations

A **bifurcation** is, heuristically, a birth and death of periodic orbits.

Recently, there are many works [AFvK⁺23], [AB25], [JKvK26] relating bifurcation with symplectic invariants, for example Conley–Zehnder indices, Floer-type Euler characteristics, etc.

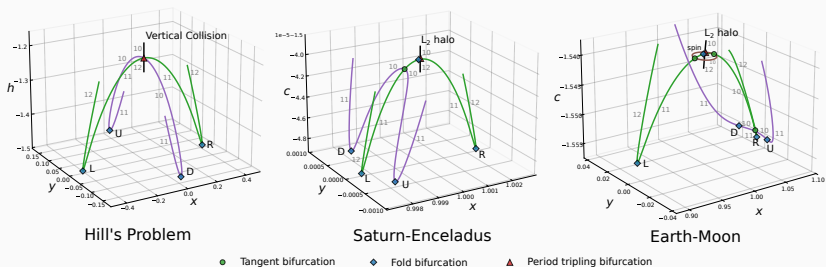


Figure 1: Bifurcation in Hill's lunar problem and CRTBP, from [JKvK26].

Classical work of Meyer [Mey70] classified generic bifurcation types.
Can we make a *symplectic dictionary* to these bifurcation types?

Motivation 2. Rotating Kepler Problem

In the rotating Kepler problem, there are three important periodic orbits: retrograde, direct and vertical collision orbits.

The multiple covers of the retrograde and direct orbits bifurcates, which produces orbits looks like a flower.

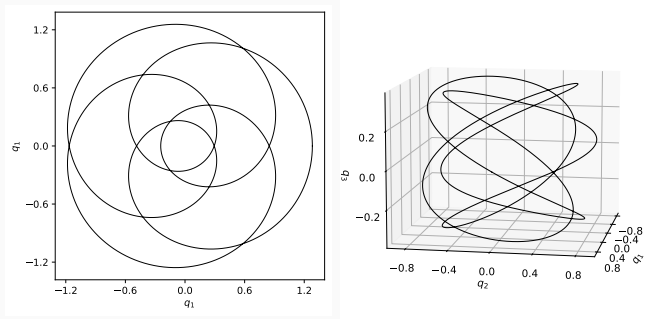
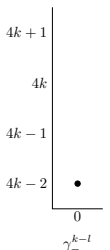
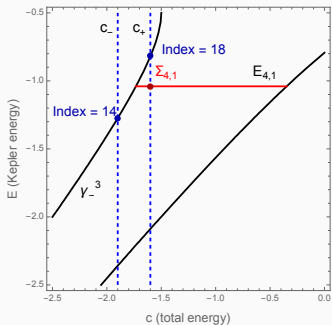


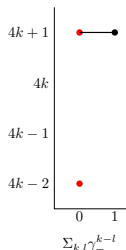
Figure 2: Resonant orbits of RKP, in courtesy of Chankyu Joung.

Motivation 2. Rotating Kepler Problem

In my previous work [Lee26], I observed this bifurcation can be translated to a change of Floer chain complex in terms of spectral sequences.



At $c_{k,l}^- - \epsilon$



At $c_{k,l}^- + \epsilon$

Slogan. Bifurcation gives a *mutation of a Floer chain complex*.

We aim to describe this effect explicitly.

Preliminaries

Bifurcation

Let $H_\varepsilon : W^4 \rightarrow \mathbb{R}$ be a 1-parameter family of Hamiltonians, and let γ_0 be a periodic orbit of H_0 with Floquet multiplier λ_0 , an eigenvalue of the linearized return map on the Poincaré section.

If $\lambda_0 \neq 1$, so the orbit is **non-degenerate**, then nearby periodic orbits form an *orbit cylinder*

$$\Gamma = \{(\gamma_\varepsilon, \varepsilon) : \gamma_\varepsilon \text{ is a periodic orbit of } H_\varepsilon\}.$$

This is a *continuation* of the orbit along bifurcation parameter.

If $\lambda_0 = 1$, so the orbit degenerates, our orbit cylinder may break:

- periodic orbits may be born or vanish, called **bifurcation**,
- if $\lambda \neq 1$ but $\lambda^k = 1$, then the k -th cover bifurcates.

Example. Period Doubling

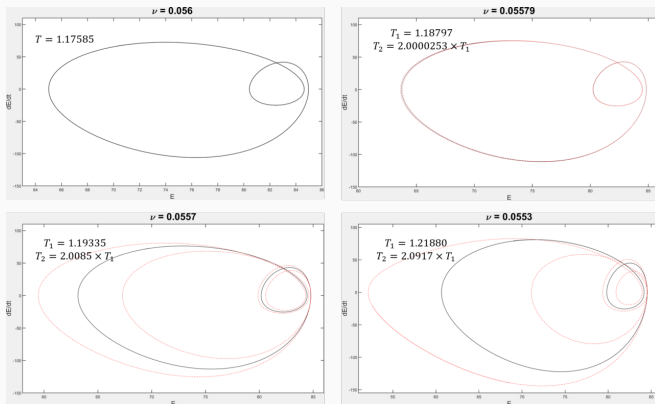


Figure 3: Period doubling in Kuramoto–Sivashinsky equation, from [Max21].

Period doubling is a generic bifurcation of double cover.

The red-dashed orbit is the new orbit emerged from the bifurcation.

Conley–Zehnder Index

The **Conley–Zehnder index** $\mu_{CZ}(\gamma)$ is an integer associated to a periodic orbit, might be understood as a *symplectic winding number*.

Our interest is a change of index along the parameter, $\mu_{CZ}(\gamma_\epsilon)$.

Along a non-degenerate family, the CZ index is constant.

Key idea. Change of μ_{CZ} = Degeneracy of γ_ϵ = Bifurcation occurs

Hence, a bifurcation can be detected via CZ index. (cf. [AB25], etc.).

Floer Homology

A **Floer homology** $HF(W, H)$ is a topological invariant associated to a symplectic manifold W and Hamiltonian H . (See [AD14] for details.)

This is determined by **Floer chain complex** $CF(W, H)$, whose

- generators are 1-periodic orbits of H ,
- grading is given by CZ indices,
- differential is determined by a count of **Floer cylinder**, whose boundaries are periodic orbits.

Through a bifurcation, $CF(W, H)$ changes a lot;

- orbits are born or vanish $\Rightarrow$ number of generators changes.
- CZ index changes $\Rightarrow$ grading changes

However, the homology doesn't change, so the differential is in charge to make the homology consistent. *Can we describe this explicitly?*

Classification of Bifurcations

Meyer's Classification

Generic Hamiltonian bifurcations are classified in [Mey70].

They are organized by the covering number k of the orbit.

- $k = 1$: birth-death
- $k = 2$: period doubling, murder (opposite stability type)
- $k = 3, 4$: phantom kiss
- $k \geq 4$: emission

Aside. We will use names from [AM78].

There are 3 extra types given, which include critical points.

Other names: touch-and-go, period tripling/quadrupling, etc.

Method of Classification

We find the periodic points of the iterated return map Θ_ε^k .

This can also be done in essentially same way, by finding generating Hamiltonian of the return maps.

- Generally, we can write Θ_ε^k as a time 1-flow of time-dependent Hamiltonian G_ε .
- If $\lambda_0^k = 1$, we can discard the time dependency by a rotating frame.
- Using **Birkhoff normal form**, the generating Hamiltonian has formal Taylor expansion by $\mathbb{Z}_k$ -symmetric polynomials.
- Critical points of G_ε correspond to the periodic orbits.

Generating Hamiltonians – Birth-death

Generic single cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^3 - \varepsilon u + \dots$$

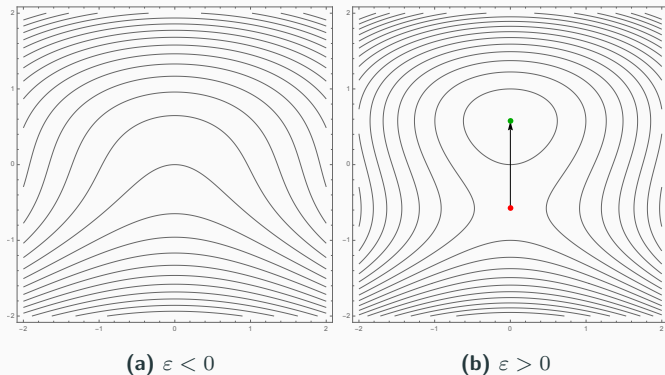


Figure 4: Critical points and gradient trajectories of birth-death bifurcation

Generating Hamiltonians – Period doubling

Generic double cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^4 - \varepsilon u + \dots$$

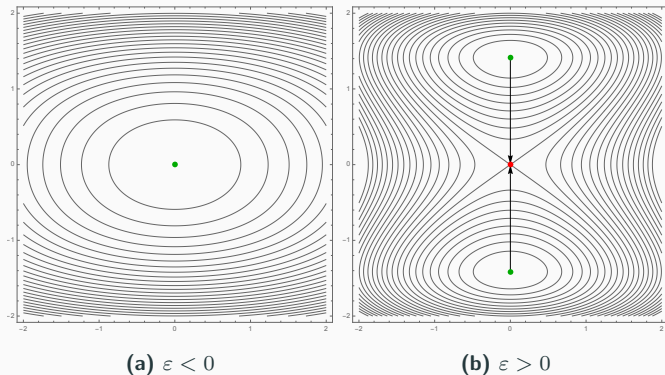


Figure 5: Period doubling bifurcation

Generating Hamiltonians – Phantom kiss

Generic 3rd or 4th cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(r, \theta) = \alpha r^k \cos k\theta - \varepsilon r^2 + \dots$$

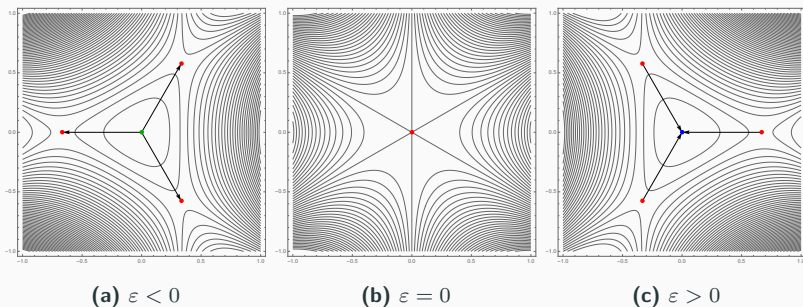


Figure 6: Phantom kiss bifurcation

Generating Hamiltonians – Emission

Generic $k \geq 4$ -th cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(r, \theta) = \alpha r^4 + \beta r^k \cos k\theta - \varepsilon r^2.$$

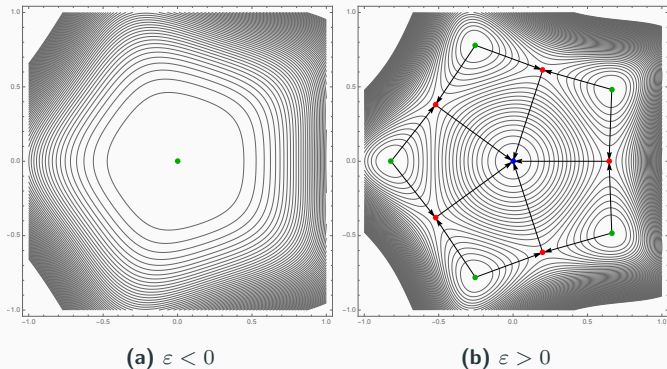


Figure 7: Emission bifurcation

Note. 4th cover is a borderline case, depending on $|\alpha|$ and $|\beta|$.

Chain Complex Mutation

Easy example. Birth-death bifurcation

Let an orbit γ_0 experiences the birth-death bifurcation along ε .

1. If $\varepsilon < 0$, there is no orbit.
2. If $\varepsilon > 0$, there are two orbits, say γ_H and γ_E .

(Very simplified) **Floer-theoretic viewpoint.**

1. If $\varepsilon < 0$, $CF = 0$.
2. If $\varepsilon > 0$, $CF = \langle \gamma_H, \gamma_E \rangle$. But both produces same homology.

Hence, $|\mu_{CZ}(\gamma_H) - \mu_{CZ}(\gamma_E)| = 1$ and they should cancel each other, so there exists a Floer cylinder between them. (cf. [GG10].)

However, the situations are more complicated for higher covers.

Main Results and Key Ideas

Main result. Full description of the mutation of Floer chain complex through 5 generic bifurcation types.

Key Ideas.

1. Construct a Hamiltonian H_ϵ which represents some bifurcation type via adiabatic approximation.
2. Gradient trajectories of G_ϵ gives Floer cylinders of H_ϵ .
3. We get cylinders connecting a simple orbit and a multiple cover.
 $\Rightarrow$ A contribution of 1 cylinder may not be exactly 1.
Using Morse–Bott cascade, we can compute the explicit chain complex, of which the resulting homology is invariant.

Adiabatic approximation of Hamiltonian

Let $\tilde{H}_\varepsilon$ be a Hamiltonian and G_ε be a generating Hamiltonian of k -th iterate of the return map.

Using **adiabatic approximation**, which is an integration along the orbit, we construct the model Hamiltonian for each bifurcation type

$$H_\varepsilon = h(r) + G_\varepsilon.$$

This Hamiltonian is not exactly same as the original one, but has the same return map and Conley–Zehnder indices with $\tilde{H}_\varepsilon$.

Morse–Bott Cascade

Our periodic orbits have S^1 -symmetry by time shift, which makes the situation slightly different from usual Floer theory.

We equip a Morse function on each orbit, and consider two critical points (maximum and minimum) as generators of a Floer chain complex.

In this case, differential is given by counting **Morse–Bott cascade**, a sequence of Morse trajectories and Floer cylinders.

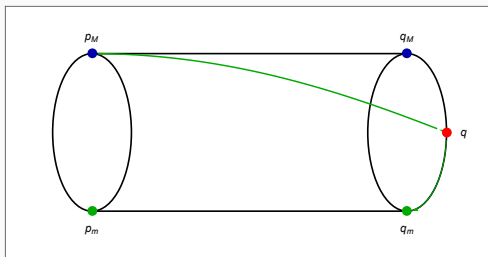


Figure 8: Morse–Bott cascade.

Chain complex mutation of Period doubling

1. If $\varepsilon < 0$, there is one orbit γ .
2. If $\varepsilon > 0$, there are two orbits; γ^2 and γ_H .
3. $\mu_{CZ}(\gamma^2)$ changes, say N to $N - 1$.
4. $\mu_{CZ}(\gamma_H)$ must be N , and there exists 1 Floer cylinder between γ^2 and γ_H , whose contribution is given as following.

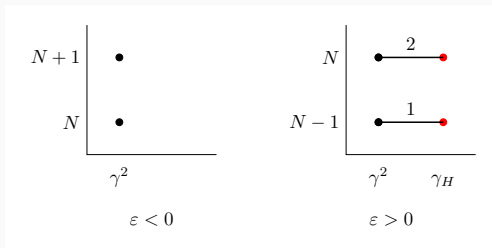


Figure 9: Chain complex mutation across the period doubling.

$\gamma_{H,\min}$ and $\gamma_{\min}^2$ cancel each other, while $\gamma_{H,\max}$ and $\gamma_{\max}^2$ don't.

Chain complex mutation of Phantom kiss and Emission

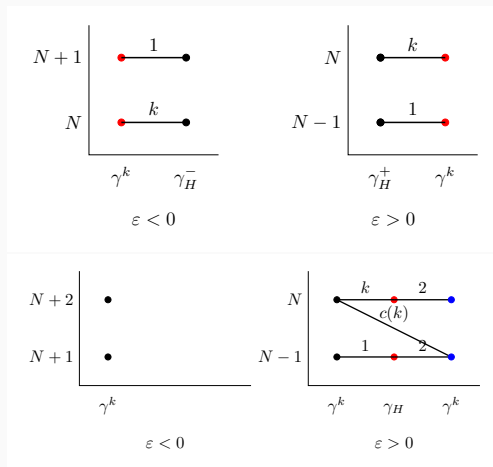


Figure 10: Chain complex mutations across phantom kiss and emission



Figure 11: Thank you for your attention!

Questions, comments, and further discussions are welcomed!



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